With India making a huge contribution to Britain's war efforts, World War I put immense strain on the railways. Railway production, of both locomotives and engines, was diverted to meet the needs of British forces outside India. At the end of the War, the Indian Railways were in a state of total disrepair. Services were downgraded, lines were ripped up to be laid elsewhere and railway workshops had formed the reputation of focussing more on ammunition and military needs than on railway matters.

When war had broken out, the Railway Board had created within itself a special war branch, which absorbed the previously created munitions branch and began dealing with the manufacture of munitions, the construction of military railways and the supply of materials and personnel to military railways in East Africa and Mesopotamia. The 'construction and working of military railways in the East' remained almost exclusively dependent on the Indian Railways system for staff and materials. Figures compiled for 1917–18 indicate that about 1,800 miles of track, 13,000 feet of bridging, 200 engines and more than 6,000 vehicles were sent out of the country. In addition to the extensive strategic use of railways to convey troops and essential military supplies like coal, strategic lines were also swiftly constructed within British India. [90]

A case in point was the army expedition at Basra in Iraq. British commander Lieutenant General Sir Frederick Stanley Maude, rebuilding his army with troops recruited from India, went by sea to Basra. While these troops were being trained, British military engineers built a field railway from the coast up to Basra and beyond. The following excerpt indicates how good infrastructure helped the war.

> Oil was vital to Britain for fuelling its most powerful warships, as well as for motor transport. British forces therefore occupied the oil fields near Basra in late 1914. A decision was then made to push up-river towards Baghdad, but this ill-fated expedition was supported by thinly stretched supply lines using river steamers and was cut off, surrounded and ultimately defeated at Kut-al-Amara.
>
> A huge effort was made to support a renewed campaign, with heavy reinforcements from India, by improving the port of Basra, building roads and railways and introducing more and better river steamers. Baghdad fell on 11 March 1917.
>
> There was also a much greater demand on the available locomotives, rolling stock and infrastructure.
>
> The adequacy of transport and supply networks played a major role in shaping strategies for operations throughout World War I and in influencing their success or failure. [91]